STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-FINITE-PROJECTIVE-MODULES-RS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 395-401 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 403-427, the complete author proof (23 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 25 of queue_examples-defos.json (pb=1085, stmt 395-401, proof 403-427).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0D3I: sources/stacks-examples-defos/pinned-original-source.tex, source lines 359–369. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06J5: sources/stacks-examples-defos/pinned-original-source.tex, source lines 427–427. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0D2J: sources/stacks-examples-defos/pinned-original-source.tex, source lines 427–427. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Example 0D3I satisfies the Rim-Schlessinger condition (RS). In particular, $\mathcal{D}ef_V$ is a deformation category for any finite dimensional vector space V over k.*

Proof. Let $A_1 \to A$ and $A_2 \to A$ be morphisms of $\mathcal{C}_\Lambda$. Assume $A_2 \to A$ is surjective. According to Formal Deformation Theory, Lemma 06J5 it suffices to show that the functor $\mathcal{F}(A_1 \times_A A_2) \to \mathcal{F}(A_1) \times_{\mathcal{F}(A)} \mathcal{F}(A_2)$ is an equivalence of categories.

Thus we have to show that the category of finite projective modules over $A_1 \times_A A_2$ is equivalent to the fibre product of the categories of finite projective modules over A_1 and A_2 over the category of finite projective modules over A. This is a special case of More on Algebra, Lemma 0D2J. We recall that the inverse functor sends

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the triple (M_1, M_2, φ) where M_1 is a finite projective A_1-module, M_2 is a finite projective A_2-module, and $\varphi : M_1 \otimes_{A_1} A \to M_2 \otimes_{A_2} A$ is an isomorphism of A-module, to the finite projective $A_1 \times_A A_2$-module $M_1 \times_\varphi M_2$. $\square$